Quiz 1

Subject: Maths1 Maths1 16 Jul 26 Total Marks: 150.00 Questions: 43 With Answers

QUESTIONS

Q1 SA *[3.00 marks]*

The zeros of the polynomial

$$p(x) = (x-2)(3x+1)q(x)$$

are $2, -\frac{1}{3}, 4, -2$, each with multiplicity 1. If the leading coefficient of $q(x)$ is 1, find $q(1)$.

ANSWER

-9 (Equal • Numeric)

Q2 SA *[4.00 marks]*

In a survey of 400 students, 220 students like Mathematics, 180 students like Physics, and 150 students like Chemistry. Further, 90 students like both Mathematics and Physics, 70 students like both Physics and Chemistry, and 60 students like both Mathematics and Chemistry. If 30 students like all three subjects, find the number of students who like none of the three subjects.

ANSWER

40 (Equal • Numeric)

Q3 SA *[3.00 marks]*

The zeros of the polynomial

$$p(x) = (x-2)(3x+1)q(x)$$

are $2, -\frac{1}{3}, 4, -2$, each with multiplicity 1. If the leading coefficient of $q(x)$ is 1, find $q(1)$.

ANSWER

-9 (Equal • Numeric)

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Q4 SA *[4.00 marks]*

A delivery company uses a prediction model to estimate the number of parcels delivered by a driver each day. For five days, the actual and predicted numbers of parcels are shown below.

Day	1	2	3	4	5
Actual Deliveries	40	52	48	60	55
Predicted Deliveries	42	50	51	x	58

The Sum of Squared Errors (SSE) for these five days is 35.

If the model underestimated the number of deliveries on Day 4, find the predicted number of deliveries on Day 4.

ANSWER

57 (Equal • Numeric)

Q5 MCQ *[2.00 marks]*

Consider the polynomial $p(x) = -2(x+3)^5(x-1)^4(x+2)^3$. Which of the following best describes the end behavior of $p(x)$?

OPTIONS

A. $p(x) \to \infty$ as $x \to \infty$ and $p(x) \to \infty$ as $x \to -\infty$

B. $p(x) \to -\infty$ as $x \to \infty$ and $p(x) \to -\infty$ as $x \to -\infty$ ✓ Correct

C. $p(x) \to \infty$ as $x \to \infty$ and $p(x) \to -\infty$ as $x \to -\infty$

D. $p(x) \to -\infty$ as $x \to \infty$ and $p(x) \to \infty$ as $x \to -\infty$

Q6 MCQ *[2.00 marks]*

Consider the polynomial $p(x) = -2(x+3)^5(x-1)^4(x+2)^3$. Which of the following best describes the end behavior of $p(x)$?

OPTIONS

A. $p(x) \to \infty$ as $x \to \infty$ and $p(x) \to \infty$ as $x \to -\infty$

B. $p(x) \to -\infty$ as $x \to \infty$ and $p(x) \to -\infty$ as $x \to -\infty$ ✓ Correct

C. $p(x) \to \infty$ as $x \to \infty$ and $p(x) \to -\infty$ as $x \to -\infty$

D. $p(x) \to -\infty$ as $x \to \infty$ and $p(x) \to \infty$ as $x \to -\infty$

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Q7 SA *[4.00 marks]*

If $x + a$ is one of the factors of

$$p(x) = x^2 + (2a - 3)x + a^2 - 2a,$$

find the value of a.

ANSWER

0 (Equal • Numeric)

Q8 MSQ *[4.00 marks]*

Which of the following statements is (are) correct?

OPTIONS

A. $y + 3 = 5(x - 2)^2$ represents a parabola whose vertex is $(2, -3)$ ✓ **Correct**

B. $p(x) = ax^5 + bx^3 + 7x - 1$, where a=0 and $b \neq 0$, then $p(x)$ is a polynomial of degree 3 ✓ **Correct**

C. The lines $4x - 8y + 12 = 0$ and $x - 2y + 3 = 0$ are coincident. ✓ **Correct**

D. The lines $2x + y - 1 = 0$ and $2x - y + 4 = 0$ are perpendicular to each other.

Q9 MSQ *[4.00 marks]*

Which of the following statements is (are) correct?

OPTIONS

A. $y + 3 = 5(x - 2)^2$ represents a parabola whose vertex is $(2, -3)$ ✓ **Correct**

B. $p(x) = ax^5 + bx^3 + 7x - 1$, where a=0 and $b \neq 0$, then $p(x)$ is a polynomial of degree 3 ✓ **Correct**

C. The lines $4x - 8y + 12 = 0$ and $x - 2y + 3 = 0$ are coincident. ✓ **Correct**

D. The lines $2x + y - 1 = 0$ and $2x - y + 4 = 0$ are perpendicular to each other.

Q10 SA *[4.00 marks]*

One root of the quadratic equation

$$x^2 - kx + 18 = 0$$

is three times the other root. If both roots are positive, find the value of k.

ANSWER

9.6 – 10 (Range • Numeric)

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Q11 MSQ *[4.00 marks]*

Let

$$p(x) = x^2 - (k+2)x + 2k.$$

Choose the correct statement(s).

OPTIONS

A. If $k = 1$, then both roots of $p(x) = 0$ are equal.

B. If $k = 2$, then both roots of $p(x) = 0$ are integers. ✔ **Correct**

C. There exists a value of k for which 0 is a root of $p(x) = 0$. ✔ **Correct**

D. If $k = 0$, then the sum of the roots of $p(x) = 0$ is 2 ✔ **Correct**

Q12 MSQ *[4.00 marks]*

Let

$$p(x) = x^2 - (k+2)x + 2k.$$

Choose the correct statement(s).

OPTIONS

A. If $k = 1$, then both roots of $p(x) = 0$ are equal.

B. If $k = 2$, then both roots of $p(x) = 0$ are integers. ✔ **Correct**

C. There exists a value of k for which 0 is a root of $p(x) = 0$. ✔ **Correct**

D. If $k = 0$, then the sum of the roots of $p(x) = 0$ is 2 ✔ **Correct**

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Q13 MSQ *[4.00 marks]*

Consider the following three straight lines:

$$\ell_1 : 2x - y + 4 = 0$$

$$\ell_2 : 4x - 2y - 3 = 0$$

$$\ell_3 : x + 2y - 5 = 0$$

Which of the following options is true?

OPTIONS

A. ℓ_1 and ℓ_2 are perpendicular.

B. ℓ_1 and ℓ_2 are parallel. ✓ **Correct**

C. ℓ_2 and ℓ_3 are parallel.

D. ℓ_1 and ℓ_3 are perpendicular. ✓ **Correct**

Q14 MSQ *[4.00 marks]*

Consider the following three straight lines:

$$\ell_1 : 2x - y + 4 = 0$$

$$\ell_2 : 4x - 2y - 3 = 0$$

$$\ell_3 : x + 2y - 5 = 0$$

Which of the following options is true?

OPTIONS

A. ℓ_1 and ℓ_2 are perpendicular.

B. ℓ_1 and ℓ_2 are parallel. ✓ **Correct**

C. ℓ_2 and ℓ_3 are parallel.

D. ℓ_1 and ℓ_3 are perpendicular. ✓ **Correct**

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Q15 MSQ [4.00 marks]

Let

$$p(x) = x^2 - (k+2)x + 2k.$$

Choose the correct statement(s).

OPTIONS

A. If $k = 1$, then both roots of $p(x) = 0$ are equal.

B. If $k = 2$, then both roots of $p(x) = 0$ are integers. ✓ Correct

C. There exists a value of k for which 0 is a root of $p(x) = 0$. ✓ Correct

D. If $k = 0$, then the sum of the roots of $p(x) = 0$ is 2 ✓ Correct

Q16 MSQ [4.00 marks]

Consider the following relations defined on the set of integers:

$$R_1 = \{(x, y) \mid x, y \in \mathbb{Z},\ y = 2x + 3\}$$

$$R_2 = \{(x, y) \mid x, y \in \mathbb{Z},\ x^2 + y^2 = 1\}$$

Choose the correct option(s):

OPTIONS

A. R_1 represents an injective function. ✓ Correct

B. R_2 represents a function.

C. $R_1 \cap R_2$ represents a function. ✓ Correct

D. R_2 represents a relation but not a function. ✓ Correct

Q17 MSQ [4.00 marks]

Let l be the equation of the line passing through the point $(2, -1)$ and perpendicular to the line

$$2x - 3y + 5 = 0.$$

Which of the following statements is (are) correct?

OPTIONS

A. The slope of l is $-\frac{3}{2}$. ✓ Correct

B. The equation of l is $3x + 2y - 4 = 0$. ✓ Correct

C. The y-intercept of l is -2.

D. The x-intercept of l is $\frac{4}{3}$. ✓ Correct

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Q18 MSQ *[4.00 marks]*

Consider the following relations defined on the set of integers:

$$R_1 = \{(x, y) \mid x, y \in \mathbb{Z},\ y = 2x + 3\}$$

$$R_2 = \{(x, y) \mid x, y \in \mathbb{Z},\ x^2 + y^2 = 1\}$$

Choose the correct option(s):

OPTIONS

A. R_1 represents an injective function. ✓ **Correct**

B. R_2 represents a function.

C. $R_1 \cap R_2$ represents a function. ✓ **Correct**

D. R_2 represents a relation but not a function. ✓ **Correct**

Q19 SA *[4.00 marks]*

One root of the quadratic equation

$$x^2 - kx + 16 = 0$$

is four times the other root. If both roots are positive, find the value of k.

ANSWER

10 (Equal • Numeric)

Q20 MSQ *[4.00 marks]*

Which of the following statements is (are) correct?

OPTIONS

A. $y - 4 = -2(x + 1)^2$ represents a parabola whose vertex is $(-1, 4)$. ✓ **Correct**

B. If $p(x) = ax^6 + bx^4 + 3x + 2$, where $a = 0$ and $b \neq 0$, then $p(x)$ is a polynomial of degree 4. ✓ **Correct**

C. The lines $3x - 6y + 27 = 0$ and $x - 2y + 9 = 0$ are coincident. ✓ **Correct**

D. The lines $x + 2y - 3 = 0$ and $2x - y + 5 = 0$ are perpendicular to each other. ✓ **Correct**

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Q21 SA *[4.00 marks]*

One root of the quadratic equation

$$x^2 - kx + 16 = 0$$

is four times the other root. If both roots are positive, find the value of k.

ANSWER

10 (Equal • Numeric)

Q22 SA *[4.00 marks]*

If $x + a$ is one of the factors of

$$p(x) = x^2 + (2a - 3)x + a^2 + 3a - 4,$$

find the value of a.

ANSWER

0 – 1 (Range • Numeric)

Q23 SA *[4.00 marks]*

If $x + a$ is one of the factors of

$$p(x) = x^2 + (2a - 3)x + a^2 + 3a - 4,$$

find the value of a.

ANSWER

0 – 1 (Range • Numeric)

Q24 SA *[3.00 marks]*

Inscript

The zeros of the polynomial

$$p(x) = (2x - 1)(x + 3)q(x)$$

are $\frac{1}{2}$, -3, 2, 5, each with multiplicity 1. If the leading coefficient of $q(x)$ is 1, find $q(1)$.

ANSWER

4 (Equal • Numeric)

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Q25 SA *[4.00 marks]*

If
$$A = \{1, 2, 3, 4, 5, 6, 7\}, \quad B = \{2, 4, 6, 8, 10\}, \quad C = \{1, 3, 5, 7, 9\},$$
find the cardinality of
$$\big((A \cup B) \cap (A \cup C)\big) - (B \cap C).$$

ANSWER

7 (Equal • Numeric)

Q26 SA *[4.00 marks]*

If
$$A = \{1, 2, 3, 4, 5, 6, 7\}, \quad B = \{2, 4, 6, 8, 10\}, \quad C = \{1, 3, 5, 7, 9\},$$
find the cardinality of
$$\big((A \cup B) \cap (A \cup C)\big) - (B \cap C).$$

ANSWER

7 (Equal • Numeric)

Q27 SA *[5.00 marks]*

The distance between the parallel lines $3x + 4y + c_1 = 0$ and $3x + 4y + c_2 = 0$ is 4, where $c_2 > c_1 > 0$. If the minimum distance between the point $(2, 3)$ and the line $3x + 4y + c_1 = 0$ is 6, find the value of $c_1 + c_2$.

ANSWER

44 (Equal • Numeric)

Q28 SA *[4.00 marks]*

Consider the points
$$A(0, 0), \quad B(x, y), \quad C(12, 6), \quad D(4, 1).$$
Suppose the point B divides the line segment AC internally in the ratio $k : 1$. If the distance between B and D is 5, find the positive value of k.

ANSWER

2 (Equal • Numeric)

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Q29 SA *[2.00 marks]*

CONTEXT / PARAGRAPH

Let

$$A = \{1, 2, 3, 6\}.$$

Define relations R_1 and R_2 on A by

$$R_1 = \{(a, b) \mid a \text{ divides } b\},$$

and

$$R_2 = \{(a, b) \mid a + b \text{ is even}\}.$$

Answer the given subquestions.

Find the number of elements in A that are related to themselves in R_1.

ANSWER

4 (Equal • Numeric)

Q30 SA *[4.00 marks]*

Consider the points

$$A(0, 0), \quad B(x, y), \quad C(12, 6), \quad D(4, 1).$$

Suppose the point B divides the line segment AC internally in the ratio $k : 1$. If the distance between B and D is 5, find the positive value of k.

ANSWER

2 (Equal • Numeric)

Q31 SA *[4.00 marks]*

The Cartesian product $B \times B$ has 16 elements. Two of the elements of the Cartesian product are $(3, 5)$ and $(7, 0)$. Find the sum of all the elements of the set B.

ANSWER

15 (Equal • Numeric)

Q32 SA *[4.00 marks]*

The Cartesian product $B \times B$ has 16 elements. Two of the elements of the Cartesian product are $(3, 5)$ and $(7, 0)$. Find the sum of all the elements of the set B.

ANSWER

15 (Equal • Numeric)

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Q33 SA *[2.00 marks]*

CONTEXT / PARAGRAPH

Let

$$A = \{1, 2, 3, 6\}.$$

Define relations R_1 and R_2 on A by

$$R_1 = \{(a, b) \mid a \text{ divides } b\},$$

and

$$R_2 = \{(a, b) \mid a + b \text{ is even}\}.$$

Answer the given subquestions.

Find the number of ordered pairs that belong to exactly one of the relations R_1 and R_2 .

ANSWER

5 (Equal • Numeric)

Q34 MCQ *[4.00 marks]*

A quadratic equation has roots α and β. It is given that

$$\alpha + \beta = 7$$

and

$$\alpha^2 + \beta^2 = 29.$$

Which of the following is the quadratic equation?

OPTIONS

A. $x^2 - 7x + 10 = 0$ ✓ Correct

B. $x^2 - 7x + 12 = 0$

C. $x^2 - 7x + 14 = 0$

D. $x^2 - 7x + 20 = 0$

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Q35 MSQ *[4.00 marks]*

CONTEXT / PARAGRAPH

Let
$$A = \{1, 2, 3, 6\}.$$
Define relations R_1 and R_2 on A by
$$R_1 = \{(a, b) \mid a \text{ divides } b\},$$
and
$$R_2 = \{(a, b) \mid a + b \text{ is even}\}.$$
Answer the given subquestions.

Which of the following statements is/are correct?

OPTIONS

A. R_1 is transitive and reflexive. ✓ **Correct**

B. R_2 is symmetric but not transitive.

C. R_2 is reflexive and transitive. ✓ **Correct**

D. $(3, 1) \in R_2$. ✓ **Correct**

Q36 MCQ *[4.00 marks]*

A quadratic equation has roots α and β. It is given that
$$\alpha + \beta = 7$$
and
$$\alpha^2 + \beta^2 = 29.$$
Which of the following is the quadratic equation?

OPTIONS

A. $x^2 - 7x + 10 = 0$ ✓ **Correct**

B. $x^2 - 7x + 12 = 0$

C. $x^2 - 7x + 14 = 0$

D. $x^2 - 7x + 20 = 0$

Q37 SA *[5.00 marks]*

The distance between the parallel lines $3x + 4y + c_1 = 0$ and $3x + 4y + c_2 = 0$ is 4, where $c_2 > c_1 > 0$. If the minimum distance between the point $(2, 3)$ and the line $3x + 4y + c_1 = 0$ is 6, find the value of $c_1 + c_2$.

ANSWER

44 (Equal • Numeric)

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Q38 SA *[5.00 marks]*

The distance between the parallel lines $3x + 4y + c_1 = 0$ and $3x + 4y + c_2 = 0$ is 4, where $c_2 > c_1 > 0$. If the minimum distance between the point $(2, 3)$ and the line $3x + 4y + c_1 = 0$ is 6, find the value of $c_1 + c_2$.

ANSWER

44 (Equal • Numeric)

Q39 SA *[2.00 marks]*

CONTEXT / PARAGRAPH

Consider the polynomial

$$p(x) = 3(x-2)^4(x+1)^7(x-5)^2.$$

Answer the given subquestions

What is the degree of $p(x)$?

ANSWER

13 (Equal • Numeric)

Q40 SA *[2.00 marks]*

CONTEXT / PARAGRAPH

Consider the polynomial

$$p(x) = 3(x-2)^4(x+1)^7(x-5)^2.$$

Answer the given subquestions

At how many of its real zeros does the graph of $p(x)$ cross the x -axis?

ANSWER

1 (Equal • Numeric)

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Q41 MCQ *[2.00 marks]*

CONTEXT / PARAGRAPH

Consider the polynomial

$$p(x) = 3(x-2)^4(x+1)^7(x-5)^2.$$

Answer the given subquestions

Which of the following best describes the end behavior of $p(x)$?

OPTIONS

A. $p(x) \to \infty$ as $x \to \infty$ and $p(x) \to \infty$ as $x \to -\infty$

B. $p(x) \to -\infty$ as $x \to \infty$ and $p(x) \to -\infty$ as $x \to -\infty$

C. $p(x) \to \infty$ as $x \to \infty$ and $p(x) \to -\infty$ as $x \to -\infty$ ✓ **Correct**

D. $p(x) \to -\infty$ as $x \to \infty$ and $p(x) \to \infty$ as $x \to -\infty$

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